

QUENCH

MIDI SEQUENCER

User Guide

Mindburner MIDI Products · Hand-built in Belfast

Firmware V3_44

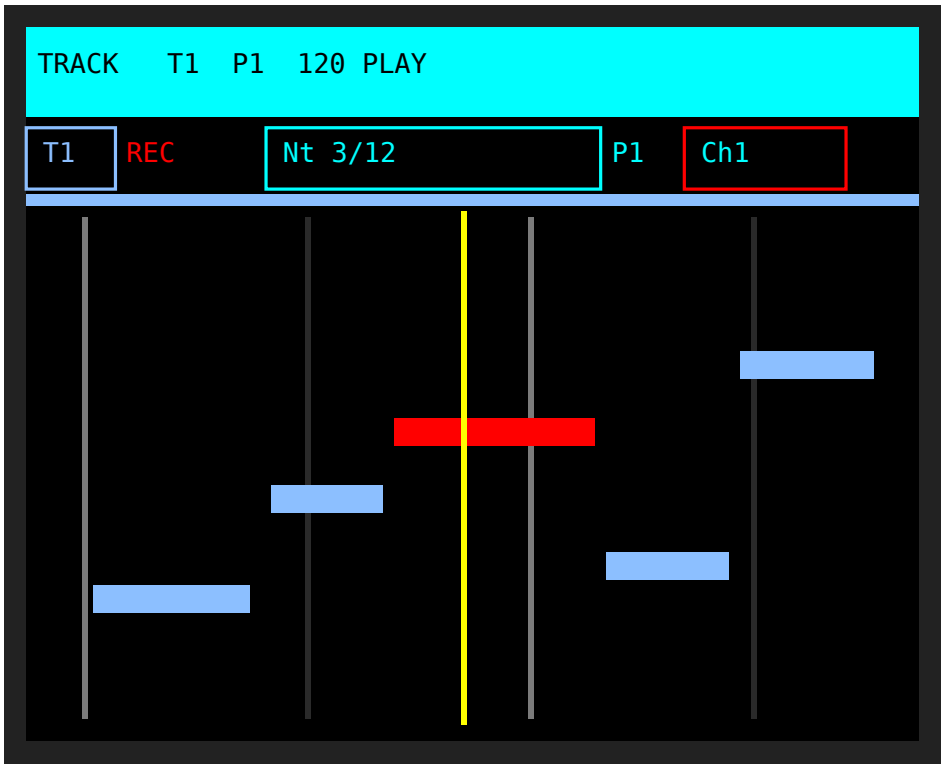
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1 Welcome to Quench

Quench is a MIDI sequencer built around one idea: **plug in and record**. Connect a MIDI keyboard to the input and your synths to the output, press play, arm the recorder and play. Whatever you play is captured with its real timing and appears on a piano roll, with the bar and beat lines drawn in so you can see exactly where you are in the loop.

There are four tracks, each with its own MIDI channel, and six patterns. Tracks 1–3 are **free tracks** for live recording. Track 4 is a **step track**, a classic step sequencer for a repeating bass line, a drum part or a filter sweep. Everything is shown on the 160 × 128 colour screen and is driven by six buttons (A to F) and one push-encoder.



Free track screen: the piano roll, the header, and the playhead (yellow).

> **TIP** Every page has a built-in help screen. Hold the encoder for two seconds on a free track, on a Track 4 page, or on the Song page and Quench lists exactly what each control does on that screen.

2 Quick start

Connect

- **MIDI IN**: your keyboard or controller. Its notes are passed straight through to the selected track's MIDI channel, so you hear the synth you are playing.
- **MIDI OUT**: your synth or synths. Use the channel numbers to send each track to a different instrument.
- Power it up. The Mindburner start-up animation plays, and your last saved song loads automatically.

Record a take in eight steps

1. You start on **Track 1**, Pattern 1, on the piano roll. To use another track turn the encoder (the **T** box in the header is lit).
2. Click the encoder until the red box sits round **Ch** and turn to choose the MIDI channel. See section 6 for the click order.
3. Press **D** to start playing. The yellow playhead sweeps across the loop.
4. Hold **C** to arm recording. The header shows **REC** in red.
5. Play your keyboard. Notes appear on the roll as you play them.
6. Hold **C** again to stop recording. Your take keeps looping.
7. To save the song, hold **D**, and when the screen says "D to SAVE" hold **D** again.
8. Press **F** at any time to jump home: Track 1, Pattern 1, with every armed state cancelled.

> **TIP** Made a mess? Hold **F** for just under a second to clear the whole take on the current track. REC stays armed, ready for another go.

3 The controls

The same buttons do the same kind of job on every page. This table is the overview; the later sections give the detail for each page.

Control	What it does
A	Short press: previous pattern. On the piano roll with a note picked, previous note. Hold: **mute** the track.
B	Short press: next pattern, or next note when a note is picked. Hold: **solo** the track.
C	Short press: next page. On a picked note in Edit, switch which value the encoder changes. Hold: the page's main action (arm REC, MIDI learn, set all, CC shape).
D	Short press: play / stop. Hold twice: save. On the Song page, hold: Save / Load screen.
E	Short press: add a note (free tracks), or clear the step (Track 4). Hold: delete the picked note, or cancel a note you are placing.
F	Tap: home, Track 1, Pattern 1, and cancel everything. Hold ¾ second: clear the track.
Encoder turn	Moves whatever the red or cyan box in the header is on: track, note, value, channel.
Encoder click	Moves the box along (see sections 6 and 7), and fixes a value you are editing.
Encoder hold	Two seconds: help screen. On a Track 4 page, a shorter hold moves the box to the track. Hold + turn on a free track: scroll the piano roll pitch.

The header bar

The coloured bar at the top of every page reads, from left to right: the **page name** (TRACK, STEP, SONG, VELOCITY, LENGTH, CC), the **track** (T1–T4), the **pattern** (P1–P6), the **tempo**, and PLAY or STOP. A red **R** means recording is armed. The small dot at the far right flashes when MIDI arrives. A thin red strip down the left edge means there are changes you have not saved yet.

> **TIP** When a flash message appears ("Cancelled", "Note added", "REC ARMED") it confirms what you just did. They are kept short on purpose.

4 Tracks and patterns

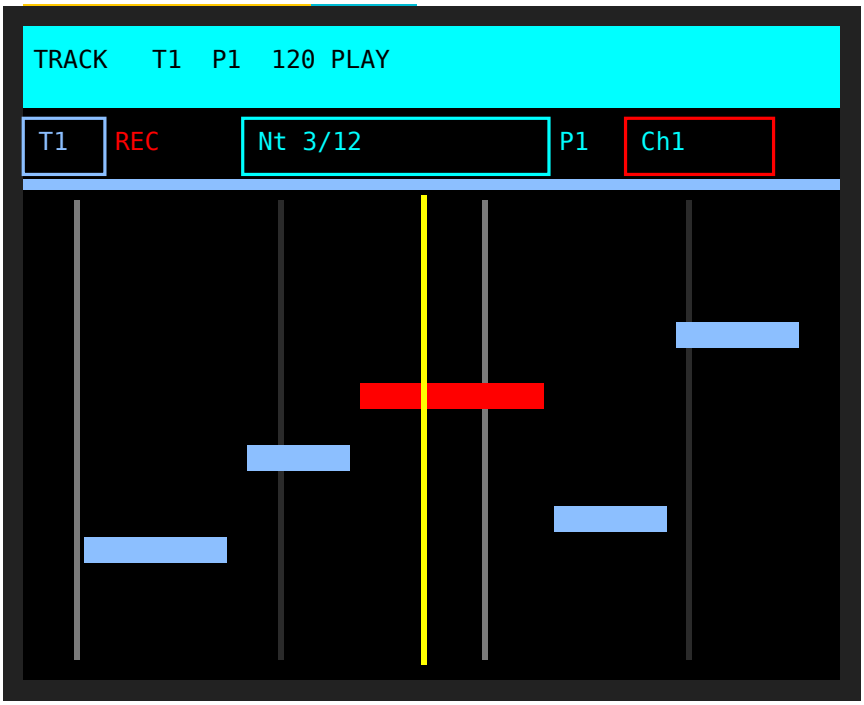
Track	Type	Use it for
1, 2, 3	Free (live record)	Playing parts in with a keyboard: melodies, chords, bass. Real timing, up to 48 notes per track per pattern. Shown as a piano roll.
4	Step	A repeating step pattern with per-step note, velocity, length and a MIDI CC value. Up to four-note chords per step.

Each track has its **own MIDI channel** (1–16), which is shared by all six patterns. There are **six patterns** (P1–P6). A pattern holds a take for each free track and a step pattern for Track 4. A and B step through the patterns, and the Song page chains them into a song.

All patterns share the same length, set in Settings (8, 12, 16, 20, 24 or 32 steps, steps being sixteenth notes) and the same time signature.

> **TIP** Mute (A hold) and solo (B hold) work on every track. The M and S marks show in the header area when they are on.

5 Recording on Tracks 1–3



Header, left to right: track · REC · the note or value field · pattern · channel · M / S.

Arm, play, record

- Press **D** to play. Recording only captures notes while the sequencer is playing.
- Hold **C** to arm. The header shows REC in red. Hold **C** again to disarm.
- Play. Notes land exactly where you played them, with their real length and velocity. Chords work: several notes at once are recorded as a chord.
- The roll draws held notes growing as you hold them, and the bar lines (brighter) and beat lines (dimmer) show the grid.
- While recording, the loop always plays forward, even if the pattern's Play Mode is Reverse.

Overdub or replace

Settings has a **Rec Mode** row. **OVERDUB** (the default) adds whatever you play to the notes already there, so you can build layers. **REPLACE** wipes the track's take when you arm; because that is destructive, with notes present you have to hold **C** a second time ("Hold C again").

Auto-quantise

Settings has an **AutoQ Free trks** row: OFF, 1/16, 1/32 or 1/64. When on, the start of each note you record is snapped to that grid. Notes you place by hand with **E** also move in steps of the same grid.

Clearing and limits

- **F hold** clears the whole take on the current track. REC stays armed.
- Each track holds 48 notes per pattern. "Take full!" means there is no room for more.
- Hold **E** deletes just the picked note (section 6).

> **TIP** Reverse and Zigzag (Song page, Play Mode) play the free tracks backwards too. With Reverse, every note sounds from where it used to end, so the whole pattern mirrors.

6 Editing notes: the header ring

On a free track the encoder always acts on the box in the header. Click moves the box to the next place. Turning then changes what the box is on. The order is:

Click	Box on	Turn does
(start)	T (track number)	Switches track, 1 to 4.
1	Note (“Nt 3/12”)	Picks a note. A and B also step through notes.
2	Edit (red)	Changes the picked note’s value; C chooses which one (below).
3	View (only when notes are wide apart)	Scrolls the pitch window up and down.
4	Ch (red)	Changes the track’s MIDI channel, 1–16.
5	back to T	Turn switches track again.

Note work comes first because it is what you do most. With no notes yet, the first click skips straight to View or Ch.

What Edit changes

In Edit, press **C** to cycle the value the encoder changes: **Pitch** → **Len** (length) → **Start** → **Vel** (velocity) → **Del** (delete) and back. At Del, a click deletes the note.

Adding a note by hand

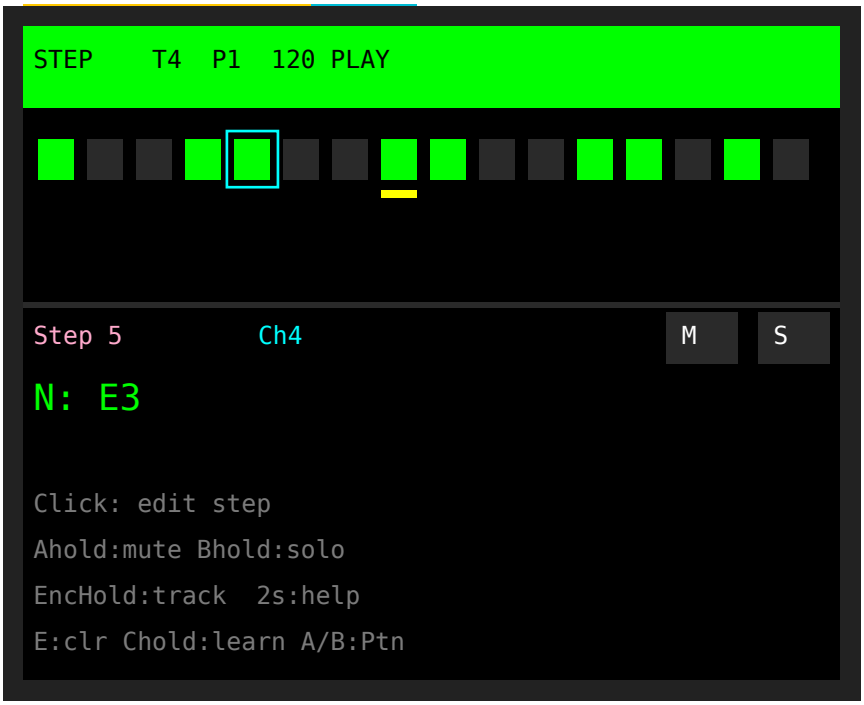
1. Press **E**. The header shows **Place:** and a position counter. Turn the encoder to slide the position along the grid.
2. Press **E** or click to fix it. The new note starts at the pitch of the previous note and drops straight into Pitch editing.
3. Changed your mind before fixing? **Hold E**: the placement is cancelled and nothing changes.
4. Made one by mistake after fixing? **Hold E** deletes the picked note.

Scrolling the pitch window

When a take is wider than the screen shows (say a low C1 and a high C6), turn to the **View** box, or simply **hold the encoder and turn** at any time.

> **TIP** Free track notes can be muted, soloed, changed in length, moved in time, and have their velocity set note by note. Nothing is permanent until you save.

7 Track 4: the step track



Track 4’s STEP page: sixteen steps, a cursor, and the hints below.

Select Track 4 by turning the encoder on the T box. It opens directly on its step grid. The grid has one square per step; lit squares carry a note, the cyan box is the cursor, and a small yellow line under a square is the playhead.

Pages (press C to move on)

Page	Shows	Edit it with
STEP	The notes: one pitch per step (chords too).	Turn moves the cursor. Click “readies” the step (red). Turn then changes the pitch. Click fixes. **E** clears the step.
SONG	See section 8.	
VELOCITY	Bars: how hard each note is struck, 0–127.	Same ready/turn/fix scheme. The cursor only stops on steps with notes. A velocity of 0 clears the step. **C hold** sets every note to this velocity.
LENGTH	Bars: how long each note holds.	Values to 100% are a share of one step; above 100% a note stretches over several steps. **C hold** sets all lengths.
CC	Track 4 only. One MIDI CC number, with a level 0–127 for every step.	Same scheme. **C hold** opens Shape: Flat, Ramp Up, Ramp Down, Triangle or Seesaw. **E hold** locks the CC number so turning changes it; E hold again to leave.

TIP The CC page belongs to **Track 4 only**. The value is sent on Track 4’s MIDI channel at every step, which makes it the way to sweep a filter or any other parameter on the synth Track 4 is playing. **Tracks 1–3 send no CC at all**: they replay only the notes you recorded, and the CC page is not available while one of them is selected. Set the CC number to NONE to switch Track 4’s CC off.

Chords

A step can hold up to **four notes**. Enter them with MIDI learn (below): notes you play within about a tenth of a second of each other stack into one chord on the same step. The STEP readout shows the root note and “x3” for a three-note chord.

MIDI learn: playing in the notes

- **C hold** on the STEP page starts listening. The screen shows “LISTENING”.
- **Stopped:** each note you play fills the step under the cursor and moves on to the next.
- **Playing:** notes land on the nearest step as the pattern plays.
- Press **C** to stop listening. **D** still starts and stops the sequencer.

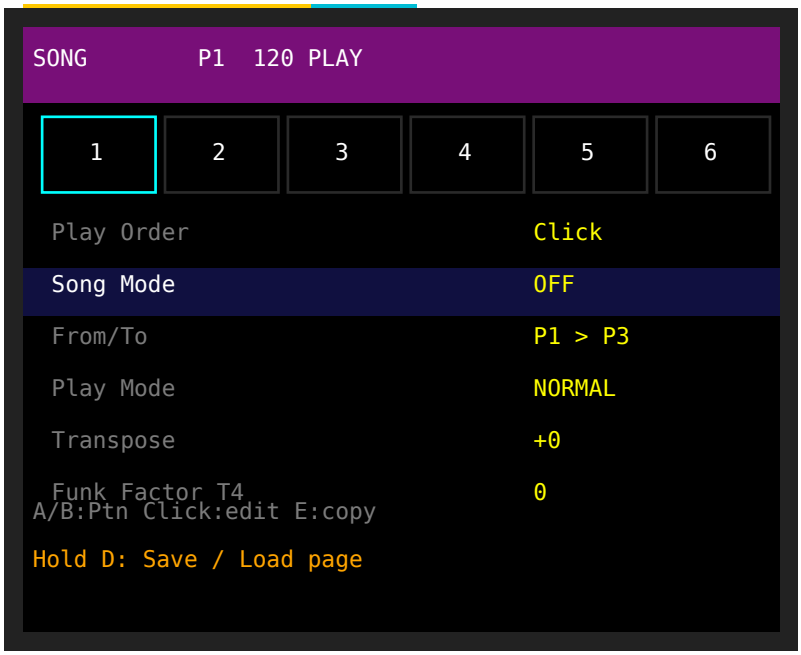
The track box on Track 4

Track 4’s pages have the same kind of header ring as the free tracks, but the step cursor is the normal state, so the box starts out off:

1. **Hold the encoder** (about half a second, without turning). A red box appears round **T4**. Turn to go to another track.
2. **Click**: the box moves to **Ch**. Turn to change the MIDI channel, 1–16.
3. **Click** again: back to the steps.

A hold and **B hold** mute and solo Track 4. **F hold** clears the whole step track. **A/B** step through patterns (while a step is readied, the encoder is busy, so A/B wait).

8 The Song page



Press C to reach it from any track. A and B choose the pattern; turn to move between rows.

Row	What it does
Pattern boxes	The six boxes at the top are the patterns. A green tint means that pattern has something recorded in it; a plain box with a grey number is empty.
Play Order	Six slots, each of which can hold any pattern (repeats allowed, e.g. 1, 1, 2, 3). Click to enter, click again to grab a slot, turn to choose its pattern, click to fix.
Song Mode	**OFF:** D plays the pattern you are viewing, looping. A and B change what you hear. **ON:** D plays From → To of the Play Order and repeats; the screen follows the song.
From / To	The slice of Play Order that plays when Song Mode is on. Click to edit From, click for To, click to finish. From higher than To wraps round: From 5 To 2 plays 5, 6, 1, 2.
Play Mode	Per pattern: NORMAL, REVERSE, RANDOM or Z (zigzag, forwards then backwards). Reverse and Z also apply to Tracks 1–3. Random only affects Track 4.
Transpose	Per pattern, −12 to +12 semitones, applied when playing. Your notes are never changed, so 0 restores the original. Skipped on MIDI channel 10 (drums). Copying a pattern copies its transpose.
Funk Factor T4	Swing, 0–10, on Track 4 only. Tracks 1–3 replay your own timing exactly as played.

To chain P1 + P2 + P3

1. On Play Order, set positions 1, 2, 3 to patterns 1, 2, 3.
2. Turn Song Mode **ON**.
3. Set From = 1 and To = 3.
4. Press **D**.

Copying a pattern

On the Song page, press **E** to open the **Copy Pattern** screen. Turn (or A / B) to move along the six patterns. **C** or a click picks the source, then pick the destination and confirm with a click: the whole pattern is copied (free-track takes, Track 4 steps, Play Mode and Transpose). **D** clears an entire pattern (confirm with D). **Hold the encoder** to clear only the current track within that pattern, then click to confirm. **E** closes the screen and takes you back to the Song page.

Save and load

Hold **D** on the Song page to open the **Save / Load** screen. Two little diskettes are shown, **Song 1** and **Song 2** (green when something is saved on them, grey when empty); **turn the encoder** to pick one. **C** saves into the chosen song (it asks you to confirm with C again), **D** loads it (it asks to confirm with D), and **E** closes. A yellow * marks the song you are working on. On every other page, hold **D** and then hold **D** again within three seconds to quick-save into that song. The song you used last loads by itself when you switch on.

> **NOTE** Saving replaces whatever is in the chosen song. Only the notes you have actually recorded are stored, so two songs normally fit easily. “Song Saved” means everything fitted; if you see a partial-save message, memory is full — clear unused takes and save again. A song saved by older firmware is converted to the smaller format automatically the first time you switch on. The Settings page shows how full the pattern memory is.

9 Settings

Settings is a boot-time menu. **Hold A during the start-up animation** (a small reminder appears late in the intro). Turn the encoder to move between rows; **click** to edit a row, turn (or A/B) to change it, then click again to finish. **Hold the encoder** for a second to save and reboot.

Row	Choices	Notes
BPM	20 – 300	Shows EXT when Sync is External.
Sync	INTERN / EXTERN	External follows MIDI clock, start and stop from the MIDI input.
Length	8, 12, 16, 20, 24, 32 steps	Shown with the bar count, e.g. 16/1b.
Time Sig	4/4, 3/4, 4/8, 5/4	Sets beat and bar markers; steps stay sixteenth notes.
AutoQ Free trks	OFF, 1/16, 1/32, 1/64	Snaps recorded note starts.
Send Clock	ON / OFF	Sends MIDI clock, start and stop to the output.
Dump MIDI	click	Sends the song as a standard MIDI file over the USB serial port.
Rec Mode	OVERDUB / REPLACE	See section 5.

The bar at the bottom shows how much of the song memory is in use: green, then yellow, then orange as it fills.

> **TIP** MIDI channels are no longer in Settings. Set each track’s channel on the track itself (section 6 and 7).

10 Help screens and quick reference

Help screens: hold the encoder for two seconds on a free track, on any Track 4 page, or on the Song page. Turn the encoder to flip pages on the free-track and Song screens; press any button to close (the screen waits until you let go). A quick double-click of the encoder also opens help.

Free track (Tracks 1–3)

Key	Action
D	Play / stop
C hold	Arm / disarm REC
C	Next value in Edit (Pitch, Len, Start, Vel, Del)
E	Add a note / fix it
E hold	Delete the picked note; cancel a placement
A / B	Previous / next pattern (note when picked)
A hold / B hold	Mute / solo
F	Home (Track 1, P1)
F hold	Clear this take
Encoder click	T → Note → Edit → View → Ch → T
Encoder hold + turn	Scroll pitch
Encoder hold 2 s	Help

Track 4 (STEP, VELOCITY, LENGTH, CC)

Key	Action
Turn	Move the step cursor
Click	Ready the step; click again to fix
E	Clear the step
C hold	STEP: MIDI learn · VEL / LEN: set all · CC: shape
C	Next page (STEP, SONG, VEL, LEN, CC)
A / B	Previous / next pattern
A hold / B hold	Mute / solo
F hold	Clear the whole track
Encoder hold	Track box; then click for Ch, click for steps
Encoder hold 2 s	Help

Song page

Key	Action
Turn	Move between rows
Click	Edit the row (click again to finish)
A / B	Previous / next pattern
D	Play / stop
D hold	Save / Load screen (Song 1 / 2)
E	Copy / clear pattern
Encoder hold 2 s	Help

11 Troubleshooting

Symptom	What to check
Nothing is recorded	The sequencer must be playing (D) and REC armed (header shows REC). The track must be a free track, 1–3.
I hear my keyboard but the wrong synth	The keyboard is passed to the **selected** track's channel. Change the track or set its Ch.
A track is silent	Look for the M (mute) or a solo on another track. A hold and B hold toggle them.
Reverse or Zigzag seems to do little	Both apply to Tracks 1–3 as well as Track 4, but Zigzag alternates direction once per pass, so on a short loop it is subtle. Random only affects Track 4. While REC is armed, playback is always forward.
The song will not chain	Song Mode must be ON, and From / To must cover the Play Order slots you set. With Song Mode OFF only the pattern you are viewing loops.
A recorded note is in the wrong place	Turn AutoQ on in Settings, or use Edit → Start to move it.
I pressed E by accident	Hold E before you fix the note to cancel, or after fixing to delete the picked note.
Lost in a menu	Press **F**: it goes home and cancels every armed state.

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